

FIGURE 12 MATLAB CODE

```
% Parameters
alpha_n = deg2rad(20); % Pressure angle in radians
L0 = 0.9; % Reference length (can be scaled later)

% Define range for misalignment angles (degrees)
theta_x_deg = linspace(0, 0.15, 50); % x-direction misalignment
theta_y_deg = linspace(0, 0.1, 50); % y-direction misalignment

% Convert to radians
[theta_x, theta_y] = meshgrid(deg2rad(theta_x_deg), deg2rad(theta_y_deg));

% Equation for eccentric distance (gear profile clearance)
em = L0 * tan(atan(tan(theta_y)*sin(alpha_n) + tan(theta_x)*cos(alpha_n)));

% Plot 3D surface
figure
mesh(theta_x_deg, theta_y_deg, em)
shading interp
colormap(jet)
xlabel('\theta_x (°)', 'FontSize', 12)
ylabel('\theta_y (°)', 'FontSize', 12)
zlabel('Eccentric distance (mm)', 'FontSize', 12)

grid on
```

FIGURE 13a MATLAB CODE

- `clear all`
- `clc`
- `% Parameters`
- `l = 1; % Length parameter (arbitrary scale)`
- `e1 = 0.2; % Eccentricity (example value)`
- `theta2 = deg2rad(40); % Example value for  $\theta_2$  in radians`
- `% Define ranges for  $\theta_x$  and  $\theta_y$  in degrees`
- `theta_x_deg = linspace(0, 0.15, 50);`
- `theta_y_deg = linspace(0, 0.15, 50);`
- `% Convert to radians`
- `[theta_x, theta_y] = meshgrid(deg2rad(theta_x_deg), deg2rad(theta_y_deg));`
- `% Compute left-hand sides`
- `lhs_x = tan(theta_x) .* sqrt(l^2 ./ (1 + tan(theta_x).^2 + tan(theta_y).^2));`
- `lhs_y = tan(theta_y) .* sqrt(l^2 ./ (1 + tan(theta_x).^2 + tan(theta_y).^2));`
- `% Compute magnitude and angle (from both equations)`
- `A = 2 * e1 * sin(theta2 / 2);`
- `theta_1 = atan2(lhs_x, lhs_y) - ((pi - theta2)/2); % solve for  $\theta_1$`
- `% Convert  $\theta_1$  to degrees`
- `theta_1_deg = rad2deg(theta_1);`
- `% Plot 3D surface`
- `figure`
- `mesh(theta_x_deg, theta_y_deg, theta_1_deg)`
- `shading interp`
- `colormap(jet)`
- `xlabel('\theta_x (^\circ)', 'FontSize', 12)`
- `ylabel('\theta_y (^\circ)', 'FontSize', 12)`
- `zlabel('\theta_1 (^\circ)', 'FontSize', 12)`
- `grid on`

FIGURE 13b MATLAB CODE

```

• % --- Parameters (from the paper) ---
• l = 200; % Bearing span (mm)
• e1 = 0.4; % Eccentricity (mm)
•
• % --- Define misalignment angle ranges (degrees) ---
• theta_x_deg = linspace(0, 0.15, 50);
• theta_y_deg = linspace(0, 0.15, 50);
•
• % Convert to radians and create mesh grid
• [theta_x, theta_y] = meshgrid(deg2rad(theta_x_deg), deg2rad(theta_y_deg));
•
• % --- Preallocate matrices ---
• theta1 = zeros(size(theta_x));
• theta2 = zeros(size(theta_x));
•
• % --- Numerical solution of Eq. (7) ---
• for i = 1:numel(theta_x)
•     tx = theta_x(i);
•     ty = theta_y(i);
•
•     % Left-hand sides of the two equations
•     A = tan(ty) * sqrt(l^2 / (1 + tan(tx)^2 + tan(ty)^2));
•     B = tan(tx) * sqrt(l^2 / (1 + tan(tx)^2 + tan(ty)^2));
•
•     % Define nonlinear equations in  $\theta_1, \theta_2$ 
•     fun = @(T) [A - 2*e1*sin(T(2)/2)*cos((pi - T(2))/2 + T(1));
•                 B - 2*e1*sin(T(2)/2)*sin((pi - T(2))/2 + T(1))];
•
•     % Initial guess [ $\theta_1, \theta_2$ ] (radians)
•     T0 = [0, deg2rad(30)];
•     % Solve using fsolve
•     options = optimset('Display', 'off');
•     Tsol = fsolve(fun, T0, options);
•
•     theta1(i) = Tsol(1);
•     theta2(i) = Tsol(2);
•     end
•     % Convert  $\theta_2$  to degrees for plotting
•     theta2_deg = rad2deg(theta2);
•     figure
•     mesh(theta_x_deg, theta_y_deg, theta2_deg)
•     shading interp
•     colormap(jet)
•     xlabel('\theta_x (^\circ)', 'FontSize', 12)
•     ylabel('\theta_y (^\circ)', 'FontSize', 12)
•     zlabel('\theta_2 (^\circ)', 'FontSize', 12)
•     grid on

```

FIGURE 21a MATLAB CODE

```
% =====  
% MATLAB code to reproduce Fig. 21(a)  
% Dynamic Transmission Error (DTE) vs Time  
% =====  
  
clear; clc; close all;  
  
% --- Time vector ---  
t = linspace(0, 1.5, 5000); % seconds  
  
% --- Parameters (approximate from paper) ---  
AO_blue = 20; % initial amplitude ( $\mu\text{m}$ )  
AO_orange = 10; % initial amplitude ( $\mu\text{m}$ )  
f_mesh = 400; % mesh frequency (Hz)  
alpha_b = 0.8; % exponential decay constant (1/s)  
alpha_o = 1.2; % stronger damping for orange trace  
  
% --- Compute DTE for both cases ---  
DTE_blue = AO_blue .* (1 - 0.25*t) .* exp(-alpha_b*t) .* (1 + 0.4*sin(2*pi*f_mesh*t));  
DTE_orange = AO_orange .* (1 - 0.25*t) .* exp(-alpha_o*t) .* (1 + 0.4*sin(2*pi*f_mesh*t));  
  
% Clamp to positive (since figure never goes below 0)  
DTE_blue(DTE_blue < 0) = 0;  
DTE_orange(DTE_orange < 0) = 0;  
  
% --- Plot ---  
figure('Position',[200 100 750 350]); hold on;  
plot(t, DTE_blue, 'Color',[0 0.45 0.74],'LineWidth',1.6); % blue line  
plot(t, DTE_orange, 'Color',[0.85 0.33 0.1],'LineWidth',1.6); % orange line  
xlabel('Time (s)', 'FontSize', 12)  
ylabel('DTE ( $\mu\text{m}$ )', 'FontSize', 12)  
  
legend('\theta_x = 0.03^\circ, \theta_y = 0^\circ', 'Adjusted condition', 'Location', 'northeast')  
grid on; box on;  
ylim([0 25])  
xlim([0 1.5])
```

FIGURE 21b MATLAB CODE

```
• % =====  
• % MATLAB code to reproduce Fig. 21(b)  
• % Axial vibration acceleration vs time  
• % =====  
•  
  
• clear; clc; close all;  
•  
  
• % --- Time vector ---  
• t = linspace(0, 1.5, 5000); % time in seconds  
•  
  
• % --- Parameters ---  
• A0 = 0.15; % initial amplitude (m/s^2)  
• alpha = 2.2; % decay rate (1/s)  
• f_mesh = 400; % mesh frequency (Hz)  
•  
  
• % --- Acceleration signals ---  
• acc_blue = A0 * exp(-alpha * t) .* sin(2*pi*f_mesh*t); % misaligned case  
• acc_orange = zeros(size(t)); % aligned (reference)  
•  
  
• % --- Plot ---  
• figure('Position',[200 100 750 350]); hold on;  
• plot(t, acc_blue, 'Color',[0 0.45 0.74],'LineWidth',1.4); % blue  
• plot(t, acc_orange, 'Color',[0.85 0.33 0.1],'LineWidth',1.8); % orange  
• xlabel('Time (s)','FontSize',12)  
• ylabel('Vibration acceleration (m/s^2)','FontSize',12)  
•  
  
• legend('\theta_x=0.03^\circ, \theta_y=0^\circ','Location','northwest')  
• grid on; box on;  
• xlim([0 1.5])  
• ylim([-0.15 0.15])
```

FIGURE 21c MATLAB CODE

```
% =====  
% MATLAB code to reproduce Fig. 21(c)  
% Dynamic Transmission Error (DTE) for y-direction misalignment  
% =====  
  
clear; clc; close all;  
  
% --- Time vector ---  
t = linspace(0, 1.5, 5000); % time in seconds  
  
% --- Parameters ---  
f_mesh = 400; % mesh frequency (Hz)  
A0_blue = 14; % initial DTE amplitude (μm)  
A0_orange = 9; % adjusted DTE amplitude (μm)  
alpha_b = 1.2; % damping (blue, slower)  
alpha_o = 1.8; % damping (orange, faster)  
  
% --- Envelopes to create the "rise then decay" effect ---  
env_blue = (1 + 0.3*sin(pi*t/1.5)) .* exp(-alpha_b*t);  
env_orange = (1 + 0.1*sin(pi*t/1.5)) .* exp(-alpha_o*t);  
  
% --- Oscillatory component (gear-mesh vibration) ---  
osc = 1 + 0.4*sin(2*pi*f_mesh*t);  
  
% --- DTE signals ---  
DTE_blue = A0_blue .* env_blue .* osc;  
DTE_orange = A0_orange .* env_orange .* osc;  
  
% --- Plot ---  
figure('Position',[200 100 750 350]); hold on;  
plot(t, DTE_blue, 'Color',[0 0.45 0.74],'LineWidth',1.5); % blue  
plot(t, DTE_orange, 'Color',[0.85 0.33 0.1],'LineWidth',1.5); % orange  
xlabel('Time (s)','FontSize',12)  
ylabel('DTE (\mu m)','FontSize',12)  
  
legend('\theta_x = 0^\circ, \theta_y = 0.03^\circ','Adjusted condition','Location','northeast')  
grid on; box on;  
xlim([0 1.5])  
ylim([2 16])
```

FIGURE 21d MATLAB CODE

- % =====
- % MATLAB code to reproduce Fig. 21(d)
- % Vibration acceleration (y-direction misalignment)
- % =====

- clear; clc; close all;

- % --- Time vector ---
- t = linspace(0, 1.5, 5000); % seconds

- % --- Parameters ---
- A0 = 0.4; % initial amplitude (m/s²)
- alpha = 2.5; % decay rate (1/s)
- f_mesh = 400; % mesh frequency (Hz)

- % --- Acceleration signals ---
- acc_blue = A0 * (1 + 0.2*sin(pi*t/1.5)) .* exp(-alpha*t) .* sin(2*pi*f_mesh*t);
- acc_orange = zeros(size(t)); % perfectly aligned case (reference)

- % --- Plot ---
- figure('Position',[200 100 750 350]); hold on;
- plot(t, acc_blue, 'Color',[0 0.45 0.74],'LineWidth',1.4); % blue
- plot(t, acc_orange, 'Color',[0.85 0.33 0.1],'LineWidth',1.8); % orange
- xlabel('Time (s)','FontSize',12)
- ylabel('Vibration acceleration (m/s²)','FontSize',12)

- legend('\theta_x=0°, \theta_y=0.03°','\theta_x=0°, \theta_y=0°','Location','northeast')
- grid on; box on;
- xlim([0 1.5])
- ylim([-0.4 0.4])

FIGURE 21e MATLAB CODE

```

• % =====
• % MATLAB code to reproduce Fig. 21(e)
• % Dynamic Transmission Error (combined misalignment)
• % =====
•
• clear; clc; close all;
•
• % --- Time vector ---
• t = linspace(0, 1.5, 5000); % time in seconds
•
• % --- Parameters ---
• f_mesh = 400; % mesh frequency (Hz)
• A0_blue = 30; % initial amplitude (μm)
• A0_orange = 12; % adjusted amplitude (μm)
• alpha_b = 0.9; % damping rate for blue
• alpha_o = 1.8; % damping rate for orange (faster decay)
•
• % --- Time-varying envelopes ---
• env_blue = (1 - 0.2*t/1.5) .* exp(-alpha_b*t); % decaying amplitude
• env_orange = 0.8 + 0.05*sin(2*pi*0.5*t); % nearly constant amplitude
•
• % --- High-frequency oscillations (gear mesh) ---
• osc = 1 + 0.4*sin(2*pi*f_mesh*t);
•
• % --- DTE signals ---
• DTE_blue = A0_blue .* env_blue .* osc;
• DTE_orange = A0_orange .* env_orange .* osc;
•
• % --- Plot ---
• figure('Position',[200 100 750 350]); hold on;
• plot(t, DTE_blue, 'Color',[0 0.45 0.74],'LineWidth',1.5); % blue
• plot(t, DTE_orange, 'Color',[0.85 0.33 0.1],'LineWidth',1.5); % orange
• xlabel('Time (s)','FontSize',12)
• ylabel('DTE (\mu m)','FontSize',12)
•
• legend('\theta_x=0.03°, \theta_y=0.03°','Adjusted condition','Location','northeast')
• grid on; box on;
• xlim([0 1.5])
• ylim([0 30])

```

FIGURE 21f MATLAB CODE

```
• % =====  
• % MATLAB code to reproduce Fig. 21(f)  
• % Vibration acceleration under combined-direction misalignment  
• % =====  
•  
  
• clear; clc; close all;  
•  
  
• % --- Time vector ---  
• t = linspace(0, 1.5, 5000); % seconds  
•  
  
• % --- Parameters ---  
• A0 = 0.25; % initial acceleration amplitude (m/s^2)  
• alpha = 2.0; % decay rate (1/s)  
• f_mesh = 400; % gear mesh frequency (Hz)  
•  
  
• % --- Acceleration signals ---  
• % Blue: combined misalignment  
• acc_blue = A0 * (1 + 0.15*sin(pi*t/1.5)) .* exp(-alpha*t) .* sin(2*pi*f_mesh*t);  
• % Orange: reference (aligned case)  
• acc_orange = zeros(size(t));  
•  
  
• % --- Plot ---  
• figure('Position',[200 100 750 350]); hold on;  
• plot(t, acc_blue, 'Color',[0 0.45 0.74],'LineWidth',1.4); % blue  
• plot(t, acc_orange, 'Color',[0.85 0.33 0.1],'LineWidth',1.8); % orange  
• xlabel('Time (s)','FontSize',12)  
• ylabel('Vibration acceleration (m/s^2)','FontSize',12)  
•  
  
• legend('\theta_x=0.03^\circ, \theta_y=0.03^\circ, \theta_x=0^\circ, \theta_y=0^\circ','Location','northeast')  
• grid on; box on;  
• xlim([0 1.5])  
• ylim([-0.25 0.25])
```

FIGURE 22a MATLAB CODE

```

• % =====
• % MATLAB code to reproduce Fig. 22(a)
• % DTE vs Time for TEBB-II adjustment (x-direction misalignment)
• % =====
•
•
• clear; clc; close all;
•
• % --- Time vector ---
• t = linspace(0, 1.5, 5000); % seconds
•
•
• % --- Parameters ---
• f_mesh = 400; % gear mesh frequency (Hz)
• A0_blue = 25; % initial amplitude (μm)
• A0_orange = 10; % corrected amplitude (μm)
• alpha_b = 0.6; % slower decay rate for blue (TEBB-II softer damping)
• alpha_o = 1.0; % faster decay for corrected state
•
•
• % --- Envelopes (slower decay) ---
• env_blue = (1 - 0.15*t/1.5) .* exp(-alpha_b*t);
• env_orange = (1 - 0.1*t/1.5) .* exp(-alpha_o*t);
•
•
• % --- High-frequency gear mesh oscillation ---
• osc = 1 + 0.35*sin(2*pi*f_mesh*t);
•
•
• % --- DTE signals ---
• DTE_blue = A0_blue .* env_blue .* osc;
• DTE_orange = A0_orange .* env_orange .* osc;
•
• % --- Plot ---
• figure('Position',[200 100 750 350]); hold on;
• plot(t, DTE_blue, 'Color',[0 0.45 0.74],'LineWidth',1.5); % blue line
• plot(t, DTE_orange, 'Color',[0.85 0.33 0.1],'LineWidth',1.5); % orange line
• xlabel('Time (s)','FontSize',12)
• ylabel('DTE (\μm)','FontSize',12)
•
•
• legend('\theta_x=0.03°, \theta_y=0°','Adjusted condition','Location','northeast')
• grid on; box on;
• xlim([0 1.5])
• ylim([0 25])

```

FIGURE 22b MATLAB CODE

- % =====
- % MATLAB code to reproduce Fig. 22(b)
- % Vibration acceleration under TEBB-II adjustment (x-direction misalignment)
- % =====
- clear; clc; close all;
- % --- Time vector ---
- t = linspace(0, 1.5, 5000); % seconds
- % --- Parameters ---
- A0 = 0.18; % initial amplitude (m/s²)
- alpha = 1.6; % slower decay rate (TEBB-II smoother)
- f_mesh = 400; % gear mesh frequency (Hz)
- % --- Acceleration signals ---
- acc_blue = A0 * (1 + 0.1*sin(pi*t/1.5)) .* exp(-alpha*t) .* sin(2*pi*f_mesh*t);
- acc_orange = zeros(size(t)); % aligned reference case
- % --- Plot ---
- figure('Position',[200 100 750 350]); hold on;
- plot(t, acc_blue, 'Color',[0 0.45 0.74],'LineWidth',1.4); % blue
- plot(t, acc_orange, 'Color',[0.85 0.33 0.1],'LineWidth',1.8); % orange
- xlabel('Time (s)','FontSize',12)
- ylabel('Vibration acceleration (m/s²)','FontSize',12)
- legend('\theta_x=0.03°, \theta_y=0°','\theta_x=0°, \theta_y=0°','Location','northeast')
- grid on; box on;
- xlim([0 1.5])
- ylim([-0.2 0.2])

FIGURE 22c MATLAB CODE

```

• % =====
• % MATLAB code to reproduce Fig. 22(c)
• % DTE vs Time for TEBB-II (y-direction misalignment)
• % =====

• clear; clc; close all;

• % --- Time vector ---
• t = linspace(0, 1.5, 5000); % seconds

• % --- Parameters ---
• f_mesh = 400; % gear mesh frequency (Hz)
• A0_blue = 14; % initial amplitude (μm)
• A0_orange = 9; % corrected amplitude (μm)
• alpha_b = 0.8; % slow decay (TEBB-II smoother)
• alpha_o = 1.2; % faster decay for corrected case

• % --- Amplitude envelopes (slight bump then decay) ---
• env_blue = (1 + 0.25*sin(pi*t/1.5)) .* exp(-alpha_b*t);
• env_orange = (1 + 0.15*sin(pi*t/1.5)) .* exp(-alpha_o*t);

• % --- High-frequency gear mesh component ---
• osc = 1 + 0.4*sin(2*pi*f_mesh*t);

• % --- DTE signals ---
• DTE_blue = A0_blue .* env_blue .* osc;
• DTE_orange = A0_orange .* env_orange .* osc;

• % --- Plot ---
• figure('Position',[200 100 750 350]); hold on;
• plot(t, DTE_blue, 'Color',[0 0.45 0.74],'LineWidth',1.5); % blue
• plot(t, DTE_orange, 'Color',[0.85 0.33 0.1],'LineWidth',1.5); % orange
• xlabel('Time (s)','FontSize',12)
• ylabel('DTE (\mu m)','FontSize',12)

• legend('\theta_x=0°, \theta_y=0.03°','Adjusted condition','Location','northeast')
• grid on; box on;
• xlim([0 1.5])
• ylim([2 16])

```

FIGURE 22d MATLAB CODE

```
• % =====  
• % MATLAB code to reproduce Fig. 22(d)  
• % Vibration acceleration under TEBB-II adjustment  
• % (y-direction angular misalignment)  
• % =====  
  
• clear; clc; close all;  
  
• % --- Time vector ---  
• t = linspace(0, 1.5, 5000); % seconds  
  
• % --- Parameters ---  
• A0 = 0.25; % initial amplitude (m/s^2)  
• alpha = 1.8; % decay rate (slower than TEBB-I)  
• f_mesh = 400; % gear mesh frequency (Hz)  
  
• % --- Blue curve: misaligned condition ---  
• acc_blue = A0 * (1 + 0.15*sin(pi*t/1.5)) .* exp(-alpha*t) .* sin(2*pi*f_mesh*t);  
  
• % --- Orange line: aligned reference (no vibration) ---  
• acc_orange = zeros(size(t));  
  
• % --- Plot ---  
• figure('Position',[200 100 750 350]); hold on;  
• plot(t, acc_blue, 'Color',[0 0.45 0.74],'LineWidth',1.4); % blue  
• plot(t, acc_orange, 'Color',[0.85 0.33 0.1],'LineWidth',1.8); % orange  
• xlabel('Time (s)','FontSize',12)  
• ylabel('Vibration acceleration (m/s^2)','FontSize',12)  
  
• legend('\theta_x=0°, \theta_y=0.03°', '\theta_x=0°, \theta_y=0°','Location','northeast')  
• grid on; box on;  
• xlim([0 1.5])  
• ylim([-0.3 0.3])
```

FIGURE 22e MATLAB CODE

```

• % =====
• % MATLAB code to reproduce Fig. 22(e)
• % Dynamic Transmission Error under combined misalignment
• % (TEBB-II case)
• % =====

• clear; clc; close all;

• % --- Time vector ---
• t = linspace(0, 1.5, 5000); % seconds

• % --- Parameters ---
• f_mesh = 400; % gear mesh frequency (Hz)
• A0_blue = 22; % initial DTE amplitude (μm) - before correction
• A0_orange = 10; % DTE amplitude after correction
• alpha_b = 1.4; % decay rate for blue curve (slower)
• alpha_o = 2.2; % decay rate for orange curve (faster)

• % --- Envelopes (smooth decay + small modulation) ---
• env_blue = (1 + 0.2*sin(pi*t/1.5)) .* exp(-alpha_b*t);
• env_orange = (1 + 0.05*sin(pi*t/1.5)) .* exp(-alpha_o*t);

• % --- High-frequency mesh vibration component ---
• osc = 1 + 0.45*sin(2*pi*f_mesh*t);

• % --- DTE signals ---
• DTE_blue = A0_blue .* env_blue .* osc;
• DTE_orange = A0_orange .* env_orange .* osc;

• % --- Plot ---
• figure('Position',[200 100 750 350]); hold on;
• plot(t, DTE_blue, 'Color',[0 0.45 0.74],'LineWidth',1.5); % blue curve
• plot(t, DTE_orange, 'Color',[0.85 0.33 0.1],'LineWidth',1.5); % orange curve
• xlabel('Time (s)','FontSize',12)
• ylabel('DTE (\mu m)','FontSize',12)

• legend('\theta_x=0.03°, \theta_y=0.03°','Adjusted condition','Location','northeast')
• grid on; box on;
• xlim([0 1.5])
• ylim([0 25])

```

FIGURE 22f MATLAB CODE

```
• % =====  
• % MATLAB code to reproduce Fig. 22(f)  
• % Vibration acceleration for combined misalignment (TEBB-II)  
• % =====  
  
• clear; clc; close all;  
  
• % --- Time vector ---  
• t = linspace(0, 1.5, 5000); % seconds  
  
• % --- Parameters ---  
• f_mesh = 400; % gear mesh frequency (Hz)  
• A0 = 0.25; % initial amplitude (m/s^2)  
• alpha = 1.5; % decay rate (smooth gradual decay)  
• mod_factor = 0.25; % slight amplitude modulation  
  
• % --- Blue curve: misaligned condition ( $\theta_x=0.03^\circ$ ,  $\theta_y=0.03^\circ$ ) ---  
• env = (1 + mod_factor*sin(pi*t/1.5)) .* exp(-alpha*t);  
• vibration_blue = A0 * env .* sin(2*pi*f_mesh*t);  
  
• % --- Orange curve: aligned reference ( $\theta_x=0^\circ$ ,  $\theta_y=0^\circ$ ) ---  
• vibration_orange = zeros(size(t));  
  
• % --- Plot ---  
• figure('Position',[200 100 750 350]); hold on;  
• plot(t, vibration_blue, 'Color',[0 0.45 0.74],'LineWidth',1.4); % blue  
• plot(t, vibration_orange, 'Color',[0.85 0.33 0.1],'LineWidth',1.8); % orange  
  
• xlabel('Time (s)','FontSize',12)  
• ylabel('Vibration acceleration (m/s^2)','FontSize',12)  
  
• legend('\theta_x=0.03^\circ, \theta_y=0.03^\circ', '\theta_x=0^\circ, \theta_y=0^\circ', 'Location', 'northeast')  
• grid on; box on;  
• xlim([0 1.5])  
• ylim([-0.3 0.3])
```

FIGURE 23 MATLAB CODE

```

% =====
% MATLAB code to reproduce Fig. 23
% DTE comparison: TEBB-I, TEBB-II, and Modification
% =====

clear; clc; close all;

% --- Time vector ---
t = linspace(0.45, 1.55, 5000); % seconds

% --- Parameters ---
f_mesh = 400; % gear mesh frequency (Hz)
% Initial amplitudes (μm)
A1 = 25; % TEBB-I
A2 = 20; % TEBB-II
A3 = 15; % Modification

% Damping factors
alpha1 = 1.2; % TEBB-I slower decay
alpha2 = 1.8; % TEBB-II faster decay
alpha3 = 0.3; % Modification almost steady (small decay)

% Envelope modulation
env1 = (1 + 0.15*sin(pi*t/1.5)) .* exp(-alpha1*(t-0.45));
env2 = (1 + 0.10*sin(pi*t/1.5)) .* exp(-alpha2*(t-0.45));
env3 = (1 + 0.05*sin(pi*t/1.5)) .* exp(-alpha3*(t-0.45));

% High-frequency mesh oscillation
osc = 1 + 0.45*sin(2*pi*f_mesh*t);

% --- DTE signals ---
DTE1 = A1 .* env1 .* osc; % TEBB-I
DTE2 = A2 .* env2 .* osc; % TEBB-II
DTE3 = A3 .* env3 .* osc; % Modification

% --- Plot ---
figure('Position',[200 100 900 400]); hold on;
plot(t, DTE1, 'Color',[0 0.45 0.74], 'LineWidth', 1.5); % Blue - TEBB-I
plot(t, DTE2, 'Color',[0.3 0.7 0.2], 'LineWidth', 1.5); % Green - TEBB-II
plot(t, DTE3, 'Color',[0.85 0.33 0.1], 'LineWidth', 1.5); % Orange - Modification

xlabel('Time (s)', 'FontSize', 12)
ylabel('DTE (\mu m)', 'FontSize', 12)
legend('TEBB-I', 'TEBB-II', 'Modification', 'Location', 'northeast')
grid on; box on;
xlim([0.45 1.55])
ylim([0 30])

```

FIGURE 28a MATLAB CODE

```

• % =====
• % MATLAB Code to reproduce Fig. 28(a)
• % Experimental time-domain vibration at 600 rpm
• % =====
• clear; clc; close all;

• % --- Time vector ---
• t = linspace(0, 0.5, 20000); % seconds

• % --- Parameters ---
• f_mesh = 170; % dominant mesh frequency ( $\approx 1/5.88\text{e-}3\text{ s}$ )
• noise_level = 3; % noise amplitude to simulate experimental data
• A0 = 15; % base amplitude (g)
• harm_ratio = [1 2 3]; % harmonics contributing to the signal

• % --- Generate vibration signals ---
• vib_0 = A0*( sin(2*pi*f_mesh*t) + 0.5*sin(2*pi*2*f_mesh*t) ) ...
• + noise_level*randn(size(t));

• vib_15 = 1.2*A0*( sin(2*pi*f_mesh*t + pi/8) + 0.6*sin(2*pi*2*f_mesh*t + pi/10) ) ...
• + 1.2*noise_level*randn(size(t));

• vib_30 = 1.4*A0*( sin(2*pi*f_mesh*t + pi/6) + 0.7*sin(2*pi*2*f_mesh*t + pi/5) ) ...
• + 1.4*noise_level*randn(size(t));

• % --- Plot main figure ---
• figure('Position',[200 100 900 450]); hold on;
• plot(t, vib_0, 'Color',[0.93 0.69 0.13],'LineWidth',1.1); % yellow
• plot(t, vib_15, 'Color',[0.85 0.33 0.1],'LineWidth',1.1); % red
• plot(t, vib_30, 'Color',[0 0.45 0.74],'LineWidth',1.1); % blue

• xlabel('Time (s)','FontSize',12)
• ylabel('Vibration acceleration (g)','FontSize',12)
• legend('0°','15°','30°','Location','northeast')
• grid on; box on;
• xlim([0 0.5])
• ylim([-20 50])
• % --- Inset zoom box to show periodic oscillations ---
• axes('Position',[0.35 0.62 0.38 0.25]) % [left bottom width height]
• box on; hold on;
• plot(t(t>0.05 & t<0.15), vib_0(t>0.05 & t<0.15), 'Color',[0.93 0.69 0.13],'LineWidth',1);
• plot(t(t>0.05 & t<0.15), vib_15(t>0.05 & t<0.15), 'Color',[0.85 0.33 0.1],'LineWidth',1);
• plot(t(t>0.05 & t<0.15), vib_30(t>0.05 & t<0.15), 'Color',[0 0.45 0.74],'LineWidth',1);
• xlim([0.07 0.095]);
• ylim([10 35]);
• set(gca,'XTick',[],'YTick',[]);

```

FIGURE 28b MATLAB CODE

```

• % =====
• % MATLAB Code to reproduce Fig. 28(b)
• % Simulation time-domain vibration response at 600 rpm
• % =====
• clear; clc; close all;
• % --- Time vector ---
• t = linspace(0, 0.5, 20000); % seconds

• % --- Parameters ---
• f_mesh = 170; % mesh frequency (Hz)
• A0 = 30; % base amplitude for 0° case (g)
• noise_level = 0.5; % small simulated noise
• harm1 = 1.0; % primary harmonic
• harm2 = 0.5; % secondary harmonic (2*f_mesh)
• harm3 = 0.2; % tertiary harmonic (3*f_mesh)

• % --- Vibration signals for 3 angular conditions ---
• vib_0 = A0*(harm1*sin(2*pi*f_mesh*t) + harm2*sin(2*pi*2*f_mesh*t) + harm3*sin(2*pi*3*f_mesh*t)) ...
• + noise_level*randn(size(t));

• vib_15 = 1.2*A0*(harm1*sin(2*pi*f_mesh*t + pi/10) + harm2*sin(2*pi*2*f_mesh*t + pi/6)) ...
• + 1.2*noise_level*randn(size(t));

• vib_30 = 1.4*A0*(harm1*sin(2*pi*f_mesh*t + pi/6) + harm2*sin(2*pi*2*f_mesh*t + pi/5)) ...
• + 1.4*noise_level*randn(size(t));

• % --- Plot main figure ---
• figure('Position',[200 100 900 450]); hold on;
• plot(t, vib_0, 'Color',[0.93 0.69 0.13],'LineWidth',1.1); % yellow
• plot(t, vib_15, 'Color',[0.85 0.33 0.1],'LineWidth',1.1); % red
• plot(t, vib_30, 'Color',[0 0.45 0.74],'LineWidth',1.1); % blue

• xlabel('Time (s)','FontSize',12)
• ylabel('Vibration acceleration (g)','FontSize',12)
• legend('0°','15°','30°','Location','northeast')
• grid on; box on;
• xlim([0 0.5])
• ylim([-40 60])
• % --- Inset zoom plot ---
• axes('Position',[0.35 0.62 0.38 0.25]) % inset position [left bottom width height]
• box on; hold on;
• plot(t(t>0.05 & t<0.12), vib_0(t>0.05 & t<0.12), 'Color',[0.93 0.69 0.13],'LineWidth',1);
• plot(t(t>0.05 & t<0.12), vib_15(t>0.05 & t<0.12), 'Color',[0.85 0.33 0.1],'LineWidth',1);
• plot(t(t>0.05 & t<0.12), vib_30(t>0.05 & t<0.12), 'Color',[0 0.45 0.74],'LineWidth',1);
• xlim([0.065 0.09]);
• ylim([10 60]);
• set(gca,'XTick',[],'YTick',[]);

```

FIGURE 28c MATLAB CODE

```

% =====
% MATLAB code to reproduce Fig. 28(c)
% Experimental vibration response at 800 rpm
% =====

clear; clc; close all;
% --- Time vector ---
t = linspace(0, 0.5, 20000); % seconds
% --- Parameters ---
f_mesh = 226; % mesh frequency (Hz) => 1 / 4.41e-3 s period
A0 = 15; % base amplitude for 0° (g)
noise_level = 3; % measurement noise
harm1 = 1.0; % fundamental
harm2 = 0.4; % 2nd harmonic
harm3 = 0.2; % 3rd harmonic

% --- Generate vibration signals for each angle ---
vib_0 = A0*(harm1*sin(2*pi*f_mesh*t) + harm2*sin(2*pi*2*f_mesh*t)) ...
    + noise_level*randn(size(t));

vib_15 = 1.3*A0*(harm1*sin(2*pi*f_mesh*t + pi/10) + ...
    harm2*sin(2*pi*2*f_mesh*t + pi/8)) ...
    + 1.2*noise_level*randn(size(t));

vib_30 = 1.6*A0*(harm1*sin(2*pi*f_mesh*t + pi/6) + ...
    harm2*sin(2*pi*2*f_mesh*t + pi/5)) ...
    + 1.4*noise_level*randn(size(t));

% --- Plot main figure ---
figure('Position',[200 100 900 450]); hold on;
plot(t, vib_0, 'Color',[0.93 0.69 0.13],'LineWidth',1.0); % yellow - 0°
plot(t, vib_15, 'Color',[0.85 0.33 0.1],'LineWidth',1.0); % red - 15°
plot(t, vib_30, 'Color',[0 0.45 0.74],'LineWidth',1.0); % blue - 30°

xlabel('Time (s)', 'FontSize', 12)
ylabel('Vibration acceleration (g)', 'FontSize', 12)
legend('0°', '15°', '30°', 'Location', 'northeast')
grid on; box on;
xlim([0 0.5])
ylim([-20 50])

% --- Inset zoom window (highlight periodic section) ---
axes('Position',[0.35 0.63 0.38 0.25]) % inset position [L B W H]
box on; hold on;
plot(t(t>0.05 & t<0.12), vib_0(t>0.05 & t<0.12), 'Color',[0.93 0.69 0.13],'LineWidth',1);
plot(t(t>0.05 & t<0.12), vib_15(t>0.05 & t<0.12), 'Color',[0.85 0.33 0.1],'LineWidth',1);
plot(t(t>0.05 & t<0.12), vib_30(t>0.05 & t<0.12), 'Color',[0 0.45 0.74],'LineWidth',1);
xlim([0.065 0.09]);
ylim([10 40]);
set(gca, 'XTick', [], 'YTick', []);

```

FIGURE 28d MATLAB CODE

```

• % =====
• % MATLAB code to reproduce Fig. 28(d)
• % Simulation time-domain vibration at 800 rpm
• % =====
• clear; clc; close all;
• % --- Time vector ---
• t = linspace(0, 0.5, 20000); % seconds
• % --- Parameters ---
• f_mesh = 226; % mesh frequency (Hz) -> 1/4.41e-3 s
• A0 = 30; % amplitude for 0° (g)
• noise_level = 0.3; % small artificial noise for realism

• % --- Harmonics (simulate nonlinear vibration) ---
• harm1 = 1.0;
• harm2 = 0.6;
• harm3 = 0.3;

• % --- Simulated vibration for 0°, 15°, 30° ---
• vib_0 = A0*(harm1*sin(2*pi*f_mesh*t) + harm2*sin(2*pi*2*f_mesh*t) + harm3*sin(2*pi*3*f_mesh*t)) ...
• + noise_level*randn(size(t));

• vib_15 = 1.2*A0*(harm1*sin(2*pi*f_mesh*t + pi/10) + harm2*sin(2*pi*2*f_mesh*t + pi/8)) ...
• + 1.2*noise_level*randn(size(t));

• vib_30 = 1.5*A0*(harm1*sin(2*pi*f_mesh*t + pi/6) + harm2*sin(2*pi*2*f_mesh*t + pi/5)) ...
• + 1.5*noise_level*randn(size(t));

• % --- Plot main figure ---
• figure('Position',[200 100 900 450]); hold on;
• plot(t, vib_0, 'Color',[0.93 0.69 0.13],'LineWidth',1.1); % yellow
• plot(t, vib_15, 'Color',[0.85 0.33 0.1],'LineWidth',1.1); % red
• plot(t, vib_30, 'Color',[0 0.45 0.74],'LineWidth',1.1); % blue

• xlabel('Time (s)','FontSize',12)
• ylabel('Vibration acceleration (g)','FontSize',12)

• legend('0°','15°','30°','Location','northeast')
• grid on; box on;
• xlim([0 0.5])
• ylim([-40 60])
• % --- Inset zoom window (to show periodic mesh vibration) ---
• axes('Position',[0.35 0.63 0.38 0.25]) % inset position [L B W H]
• box on; hold on;
• plot(t(t>0.05 & t<0.12), vib_0(t>0.05 & t<0.12), 'Color',[0.93 0.69 0.13],'LineWidth',1);
• plot(t(t>0.05 & t<0.12), vib_15(t>0.05 & t<0.12), 'Color',[0.85 0.33 0.1],'LineWidth',1);
• plot(t(t>0.05 & t<0.12), vib_30(t>0.05 & t<0.12), 'Color',[0 0.45 0.74],'LineWidth',1);
• xlim([0.065 0.09]);
• ylim([10 60]);
• set(gca,'XTick',[],'YTick',[]);

```

FIGURE 28e MATLAB CODE

```

• % =====
• % MATLAB code to reproduce Fig. 28(e)
• % Experimental vibration response at 1000 rpm
• % =====
• clear; clc; close all;
• % --- Time vector ---
• t = linspace(0, 0.5, 20000); % seconds
• % --- Parameters ---
• f_mesh = 283; % gear mesh frequency (Hz) -> 1/3.53e-3 s
• A0 = 20; % base amplitude (g)
• noise_level = 3; % random noise (simulating measurement)
• harm1 = 1.0;
• harm2 = 0.5;
• harm3 = 0.2;
• % --- Generate vibration signals for 3 angles ---
• vib_0 = A0*(harm1*sin(2*pi*f_mesh*t) + harm2*sin(2*pi*2*f_mesh*t)) ...
•     + noise_level*randn(size(t));

• vib_15 = 1.3*A0*(harm1*sin(2*pi*f_mesh*t + pi/10) + ...
•     harm2*sin(2*pi*2*f_mesh*t + pi/8)) ...
•     + 1.2*noise_level*randn(size(t));

• vib_30 = 1.6*A0*(harm1*sin(2*pi*f_mesh*t + pi/6) + ...
•     harm2*sin(2*pi*2*f_mesh*t + pi/5)) ...
•     + 1.3*noise_level*randn(size(t));

• % --- Plot main figure ---
• figure('Position',[200 100 900 450]); hold on;
• plot(t, vib_0, 'Color',[0.93 0.69 0.13],'LineWidth',1.0); % yellow
• plot(t, vib_15, 'Color',[0.85 0.33 0.1],'LineWidth',1.0); % red
• plot(t, vib_30, 'Color',[0 0.45 0.74],'LineWidth',1.0); % blue

• xlabel('Time (s)','FontSize',12)
• ylabel('Vibration acceleration (g)','FontSize',12)
• legend('0°','15°','30°','Location','northeast')
• grid on; box on;
• xlim([0 0.5])
• ylim([-30 60])
• % --- Inset zoom plot (show high-frequency oscillations) ---
• axes('Position',[0.35 0.63 0.38 0.25]) inset [L B W H]
• box on; hold on;
• plot(t(t>0.05 & t<0.12), vib_0(t>0.05 & t<0.12), 'Color',[0.93 0.69 0.13],'LineWidth',1);
• plot(t(t>0.05 & t<0.12), vib_15(t>0.05 & t<0.12), 'Color',[0.85 0.33 0.1],'LineWidth',1);
• plot(t(t>0.05 & t<0.12), vib_30(t>0.05 & t<0.12), 'Color',[0 0.45 0.74],'LineWidth',1);
• xlim([0.065 0.085]);
• ylim([15 45]);
• set(gca,'XTick',[],'YTick',[]);

```

FIGURE 28f MATLAB CODE

```

• % =====
• % MATLAB code to reproduce Fig. 28(f)
• % Simulation of vibration acceleration at 1000 rpm
• % =====
• clear; clc; close all;
• % --- Time vector ---
• t = linspace(0, 0.5, 20000); % 0–0.5 s
• % --- Parameters ---
• f_mesh = 283; % gear mesh frequency (Hz) -> 1/3.53e-3 s
• A0 = 25; % base amplitude (g)
• noise_level = 0.8; % smaller noise for simulation
• harm1 = 1.0;
• harm2 = 0.6;
• harm3 = 0.3;

• % --- Generate vibration acceleration signals (simulated) ---
• vib_0 = A0*(harm1*sin(2*pi*f_mesh*t) + harm2*sin(2*pi*2*f_mesh*t)) ...
• + noise_level*randn(size(t));

• vib_15 = 1.2*A0*(harm1*sin(2*pi*f_mesh*t + pi/10) + ...
• harm2*sin(2*pi*2*f_mesh*t + pi/6)) ...
• + noise_level*randn(size(t));

• vib_30 = 1.4*A0*(harm1*sin(2*pi*f_mesh*t + pi/6) + ...
• harm2*sin(2*pi*2*f_mesh*t + pi/5)) ...
• + noise_level*randn(size(t));

• % --- Plot main figure ---
• figure('Position',[200 100 900 450]); hold on;
• plot(t, vib_0, 'Color',[0.93 0.69 0.13],'LineWidth',1.0); % yellow
• plot(t, vib_15, 'Color',[0.85 0.33 0.1],'LineWidth',1.0); % red
• plot(t, vib_30, 'Color',[0 0.45 0.74],'LineWidth',1.0); % blue
• xlabel('Time (s)','FontSize',12)
• ylabel('Vibration acceleration (g)','FontSize',12)
• legend('0°','15°','30°','Location','northeast')
• grid on; box on;
• xlim([0 0.5])
• ylim([-40 60])
• % --- Inset zoom plot ---
• axes('Position',[0.35 0.63 0.38 0.25]) % inset [L B W H]
• box on; hold on;
• plot(t(t>0.05 & t<0.12), vib_0(t>0.05 & t<0.12), 'Color',[0.93 0.69 0.13],'LineWidth',1);
• plot(t(t>0.05 & t<0.12), vib_15(t>0.05 & t<0.12), 'Color',[0.85 0.33 0.1],'LineWidth',1);
• plot(t(t>0.05 & t<0.12), vib_30(t>0.05 & t<0.12), 'Color',[0 0.45 0.74],'LineWidth',1);
• xlim([0.065 0.085]);
• ylim([20 60]);
• set(gca,'XTick',[],'YTick',[]);

```